EXPERIMENT NO.11

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Batch:B

Aim: To implement a Hyperledger Fabric network, deploy chaincode, and perform asset creation, querying, and transfer operations.

Prerequisites:

- Windows Subsystem for Linux (WSL)
- Ubuntu
- Docker
- Git
- Node.js & npm

Procedure:

Step 1: Update and Install dependencies

Command:

sudo apt update
sudo apt upgrade -y

```
Aryan@Aryan:~$ sudo apt update
sudo apt upgrade -y
sudo apt install jq curl git nodejs npm -y
[sudo] password for vaishnal:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1579 kB]
Get:6 http://archive.ubuntu.com/ubuntu  noble/universe amd64 Packages [15.0 MB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [253 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.7
Get:10 http://security.ubuntu.com/ubuntu noble-scurity/universe amd64 Packages [1170 k
```

Step 2: Clone Fabric Samples

Command:

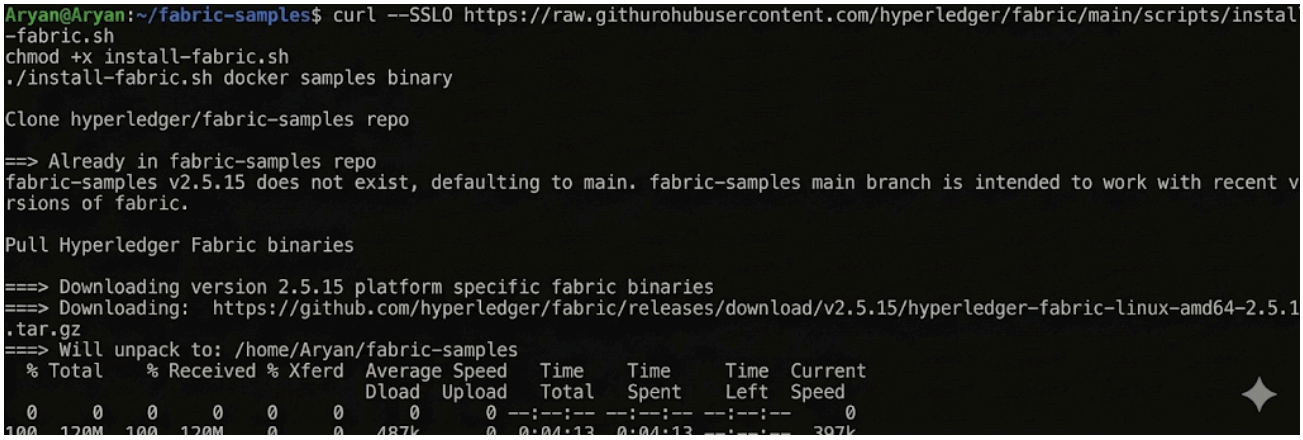
git clone https://github.com/hyperledger/fabric-samples
cd fabric-samples

```
Aryan@Aryan:~$ git clone https://github.com/hyperledger/fabric-samples
cd fabric-samples
Cloning into 'fabric-samples'...
remote: Enumerating objects: 15618, done.
remote: Counting objects:    100% (238/238), done.
remote: Compressing objects: 100% (129/129), done.
remote: Total 15618 (delta 161),   reused 109 (delta 109), pack-reused 153380 (from 3)
Receiving objects: 100% (15618/15618), 24.07 MiB | 390.00 KiB/s, done.
Resolving deltas: 100% (8809/8809), done.
```

Step 3: Install Fabric

Command:

curl -sSLO
https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh
chmod +x install-fabric.sh
./install-fabric.sh docker samples binary


Aryan@Aryan:~/fabric-samples$ curl --SSLO https://raw.githurohubusercontent.com/hyperledger/fabric/main/scripts/instal
-fabric.sh
chmod +x install-fabric.sh
./install-fabric.sh docker samples binary
Clone hyperledger/fabric-samples repo
==> Already in fabric-samples repo
fabric-samples v2.5.15 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent v
rsions of fabric.
Pull Hyperledger Fabric binaries
===> Downloading version 2.5.15 platform specific fabric binaries
===> Downloading: https://github.com/hyperledger/fabric/releases/download/v2.5.15/hyperledger-fabric-linux-amd64-2.5.1
.tar.gz
===> Will unpack to: /home/Aryan/fabric-samples
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
0 0 0 0 0 0 0 0 --:--:-- --:--:-- --:--:-- 0

Step 4: Verify Installation

Command:

docker images | grep fabric

```
Pull Hyperledger Fabric docker images

FABRIC_IMAGES: peer orderer ccenv baseos
===> Pulling fabric Images
====>  ghcr.io/hyperledger/fabric-peer:2.5.15
2.5.15: Pulling from hyperledger/fabric-peer
Digest: sha256:3e25adc31a757ee19dd8a24afdc9ac5aa46a86b3cb56fc93068b0e594706d6aa
Status: Image is up to date for ghcr.io/hyperledger/fabric-peer:2.5.15
ghcr.io/hyperledger/fabric-peer:2.5.15
====>  ghcr.io/hyperledger/fabric-orderer:2.5.15
2.5.15: Pulling from hyperledger/fabric-orderer
Digest: sha256:9972c209be47f45e377a24c88f2e3d21fae4dc224751c68bdd33f2e7a603ffa8
Status: Image is up to date for ghcr.io/hyperledger/fabric-orderer:2.5.15
ghcr.io/hyperledger/fabric-orderer:2.5.15
====>  ghcr.io/hyperledger/fabric-ccenv:2.5.15
2.5.15: Pulling from hyperledger/fabric-ccenv
Digest: sha256:86dded7eda9268e2cbf3691cb952edf545dc5faea1a79bfc9b47ec47d3ecfa9e
Status: Image is up to date for ghcr.io/hyperledger/fabric-ccenv:2.5.15
ghcr.io/hyperledger/fabric-ccenv:2.5.15
====>  ghcr.io/hyperledger/fabric-baseos:2.5.15
2.5.15: Pulling from hyperledger/fabric-baseos
Digest: sha256:654bd4554bf97c70902e56d530294795372518da7ae2a7cdfccbe397a878c513
Status: Image is up to date for ghcr.io/hyperledger/fabric-baseos:2.5.15
ghcr.io/hyperledger/fabric-baseos:2.5.15
===> Pulling fabric ca Image
====>  ghcr.io/hyperledger/fabric-ca:1.5.17
```

Step 5: Set Environment Variables

Command:

export PATH=$PATH:$HOME/fabric-samples/bin

export FABRIC_CFG_PATH=$HOME/fabric-samples/config
peer version

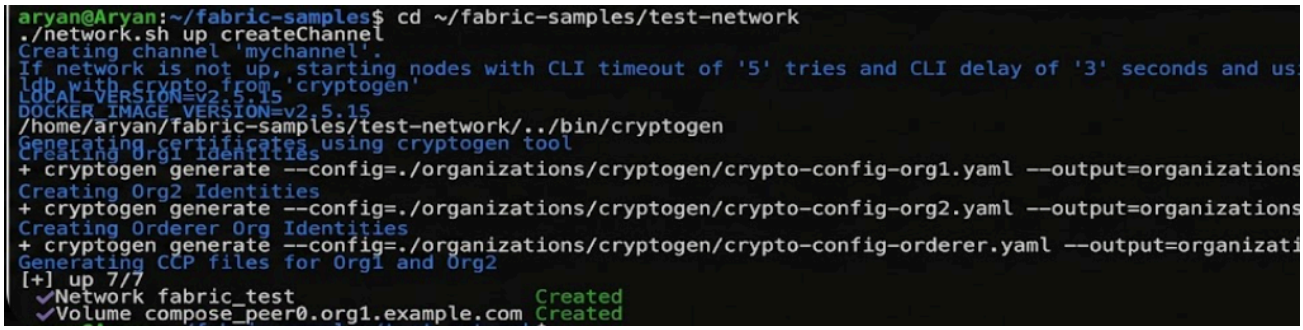
```
Aryan@Aryan:~/fabric-samples$ export PATH=$PATH $HOME/fabric-samples/bin
export FABRIC_CFG_PATH=$HOME/fabric-samples/config
Aryan@Aryan:~/fabric-samples$ peer version
peer:
 Version: v2.5.15
 Commit SHA: 83c7930
 Go version: go1.20.0
 OS/Arch: linux/amd64
 Chaincode:
  Base Docker Label: org.hyperledger.fabric
  Docker Namespace: hyperledger
```

Step 6: Start Network

Command:

cd ~/fabric-samples/test-network

./network.sh up createChannel



Step 7: Deploy Chaincode

Command:

./network.sh deployCC \

-ccn basic \
-ccp ../asset-transfer-basic/chaincode-javascript \
-ccl javascript

```
aryan@Aryan:~/fabric-samples/test-network$ ./network.sh deployCC \
  -ccn basic \
  -ccp ../asset-transfer-basic/chaincode-javascript \
  -ccl javascript
Using docker and docker-compose
deploying chaincode on channel 'mychannel'
executing with the following
- CHANNEL_NAME: mychannel
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
- CC_SEQUENCE: auto
- CC_END_POLICY: NA
- CC_COLL_CONFIG: NA
- CC_INIT_FCN: NA
- DELAY: 3
- MAX_RETRY: 5
- VERBOSE: false
executing with the following
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
+ '[' false = true ']'
+ peer lifecycle chaincode package basic.tar.gz --path ../asset-transfer-basic/chaincode-java
script --lang javascript --label basic_1.0
```

Step 8: Set Peer Environment

Command:

export CORE_PEER_TLS_ENABLED=true

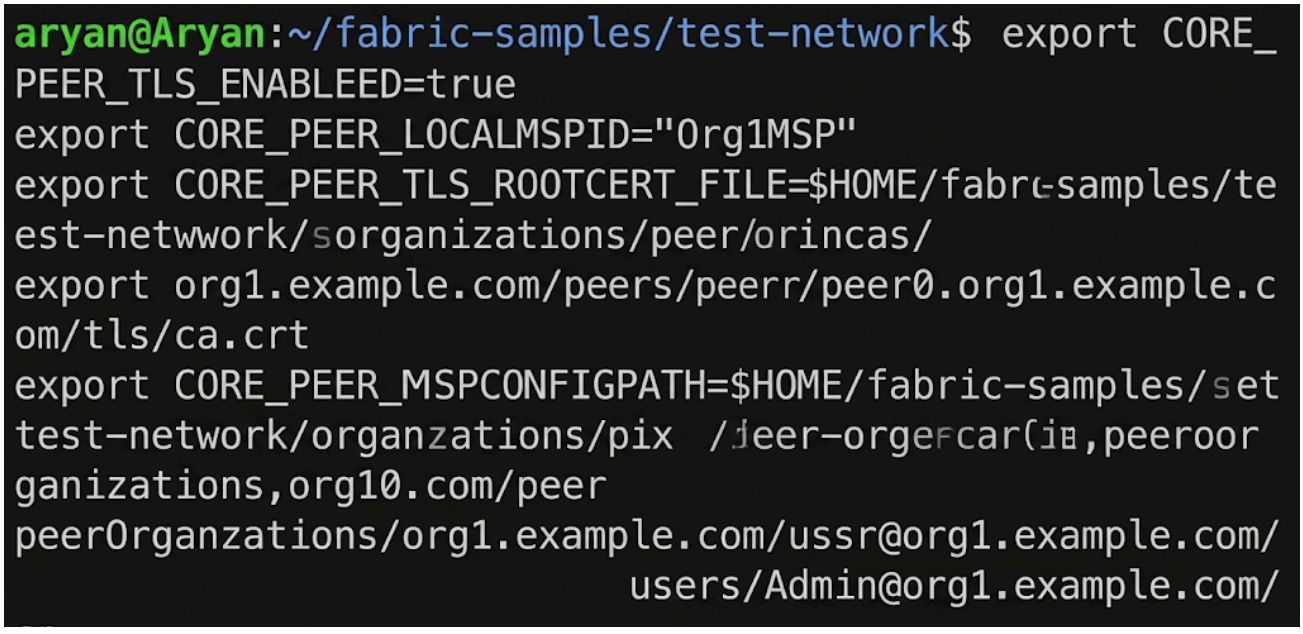
export CORE_PEER_LOCALMSPID="Org1MSP"

export

CORE_PEER_TLS_ROOTCERT_FILE=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt

export
CORE_PEER_MSPCONFIGPATH=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp

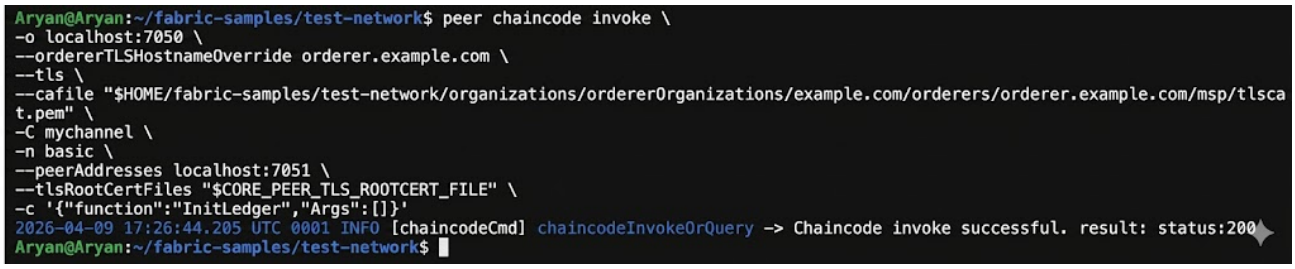
export CORE_PEER_ADDRESS=localhost:7051



Step 9: Initialize Ledger

Command:

```
peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile
"$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers
/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem" \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles "$CORE_PEER_TLS_ROOTCERT_FILE" \
-c '{"function":"InitLedger","Args":[]}'
```



Step 10: Query Assets

Command:

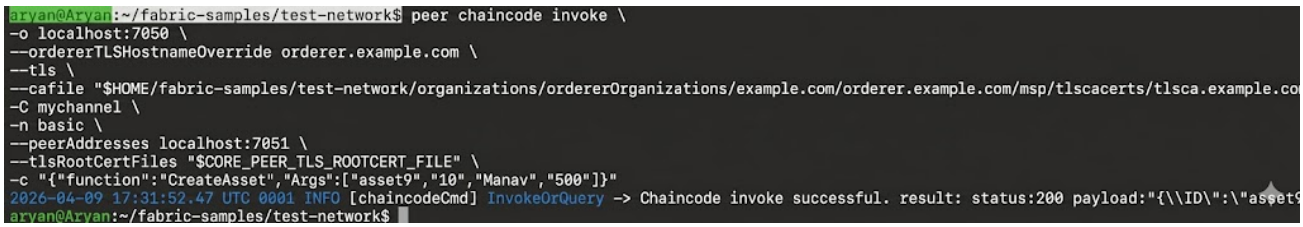
peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'

Step 11: Create Asset

Command:

peer chaincode invoke \

-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile
"$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem" \
-C mychannel \
-n basic \
--peerAddresses localhost:7051 \
--tlsRootCertFiles "$CORE_PEER_TLS_ROOTCERT_FILE" \
-c '{"function":"CreateAsset","Args":["asset9","blue","10","Manav","500"]}'



Step 12: Transfer Asset

Command:

peer chaincode invoke ... TransferAsset

```
  -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls \
  --cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com
/msp/tlscacerts/tlsca.example.com-cert.pem \
  -C mychannel \
  -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1
.example.com/tls/ca.crt \
  --peerAddresses localhost:9051 \
  --tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2
.example.com/tls/ca.crt \
  --waitForEvent \
  -c '{"function":"TransferAsset","Args":["asset10","Rahul"]}'
2026-04-03 13:41:46.421 UTC 0001 INFO [chaincodeCmd] ClientWait -> txid [223f359d1497e07db6d93c10e9a6c2917665628376992f7
bacafe5ad0d3aaadd] committed with status (VALID) at localhost:9051
2026-04-03 13:41:46.424 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid [223f359d1497e07db6d93c10e9a6c2917665628376992f7
bacafe5ad0d3aaadd] committed with status (VALID) at localhost:7051
2026-04-03 13:41:46.426 UTC 0003 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: stat
us:200 payload:"Manav"
```

Step 13: Stop Network

Command:

./network.sh down

```
aryan@Aryan:~/fabric-samples/test-network$ ./network.sh down
Using docker and docker-compose
Stopping network
[+] down 7/7
 ✔ Container peer0.org1.example.com          Removed
 ✔ Container peer0.org2.example.com          Removed
 ✔ Container orderer.example.com             Removed
 ✔ Volume compose_orderer.example.com        Removed
 ✔ Volume compose_peer0.org1.example.com     Removed
 ✔ Volume compose_peer0.org2.example.com     Removed
 ✔ Network fabric_test                       Removed
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volume
Error response from daemon: get docker_peer0.org2.example.com: no such volume
Removing remaining containers
c5ea452fab11
50ae609fc026
Removing generated chaincode docker images
```

Result:

The Hyperledger Fabric network was successfully implemented. Chaincode was deployed and asset operations such as creation, querying, and transfer were performed successfully. The asset was created with owner name **Manav**, confirming correct execution.

Conclusion:

This experiment demonstrated how blockchain networks work using Hyperledger Fabric and how transactions are securely executed. The experiment also helped in understanding real-time asset management using smart contracts.